

BIOLOGY

9747/01

Paper 1 Multiple Choice

12 September 2008

1 hour 15 min

Additional Materials: Multiple Choice Answer Sheet

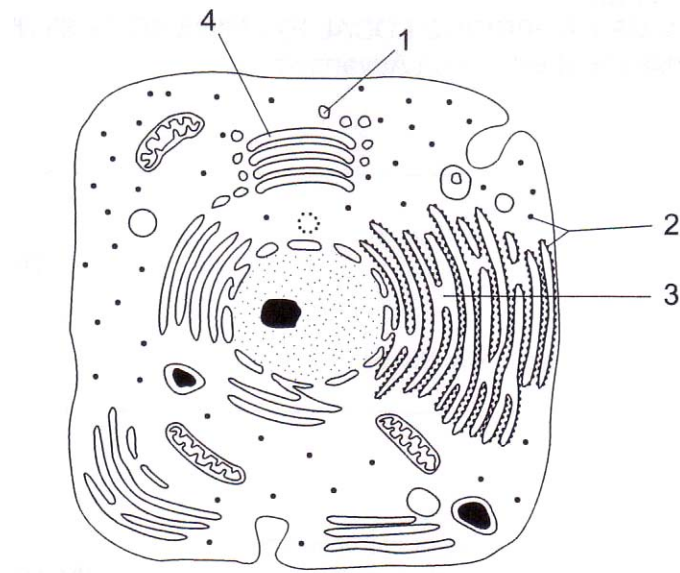
READ THESE INSTRUCTIONS FIRST

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Calculators may be used.

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1. The diagram shows the ultrastructure of an animal cell.



What are the functions of the labelled structures?

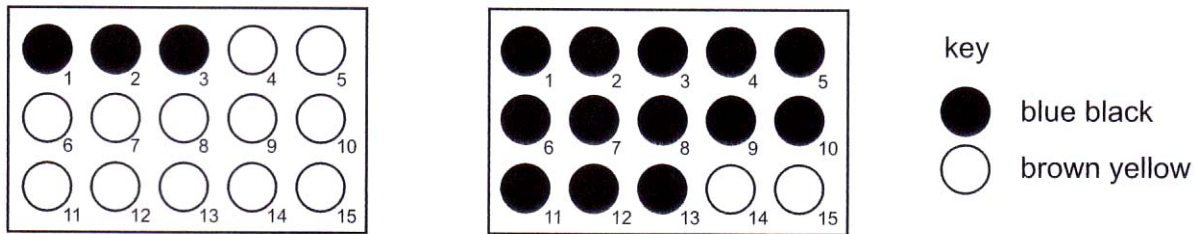
	Synthesizing proteins from amino acids	Transporting proteins	Processing proteins	Secreting digestive enzymes
A	2	1	4	3
B	1	4	2	3
C	2	3	4	1
D	1	3	2	4

2. From which cell organelle are nucleic acids absent?

- A** chloroplast
- B** Golgi apparatus
- C** mitochondrion
- D** ribosome

3

3. An experiment was carried out to investigate the digestion of starch using amylase at two different temperatures. A sample was removed from each mixture at 15 second intervals and placed onto a spotting tile well containing two drops of iodine in KI solution. The results are shown in the diagram.



Which shows the correct temperatures and times for the complete digestion of starch?

	temperature / °C	time / s
A	30 10	3.15 0.45
B	30 10	45 195
C	10 30	45 195
D	10 30	3.15 0.45

4. In active mammalian muscle, the oxidation of 1 g of fat requires 2 litres of oxygen and approximately 2 cm³ of water is produced. For each litre of oxygen used, 1 cm³ of the water produced is lost in the expired air. What is the net water gain to the body from the complete oxidation of 100 g of fat?

- A** 0
- B** 10 cm³
- C** 100 cm³
- D** 200 cm³

5. Which statement about triglycerides is correct?

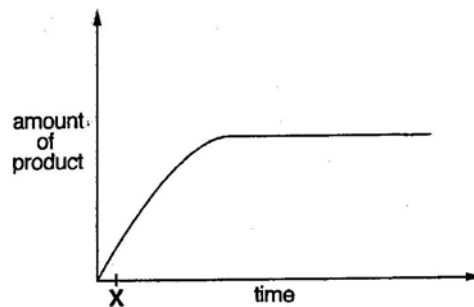
- A They are made up of three fatty acids combined with glycogen.
- B They are more saturated with hydrogen compared with phospholipids.
- C They form a bilayer in the membranes of cells.
- D They have a lower ratio of oxygen to carbon compared with carbohydrates.

6. Which sequence shows the correct order of increasing energy content of molecules?

least —————→ most

- | | | | |
|---|---------|---------|---------|
| A | ATP | starch | sucrose |
| B | ATP | sucrose | starch |
| C | sucrose | starch | ATP |
| D | starch | sucrose | ATP |

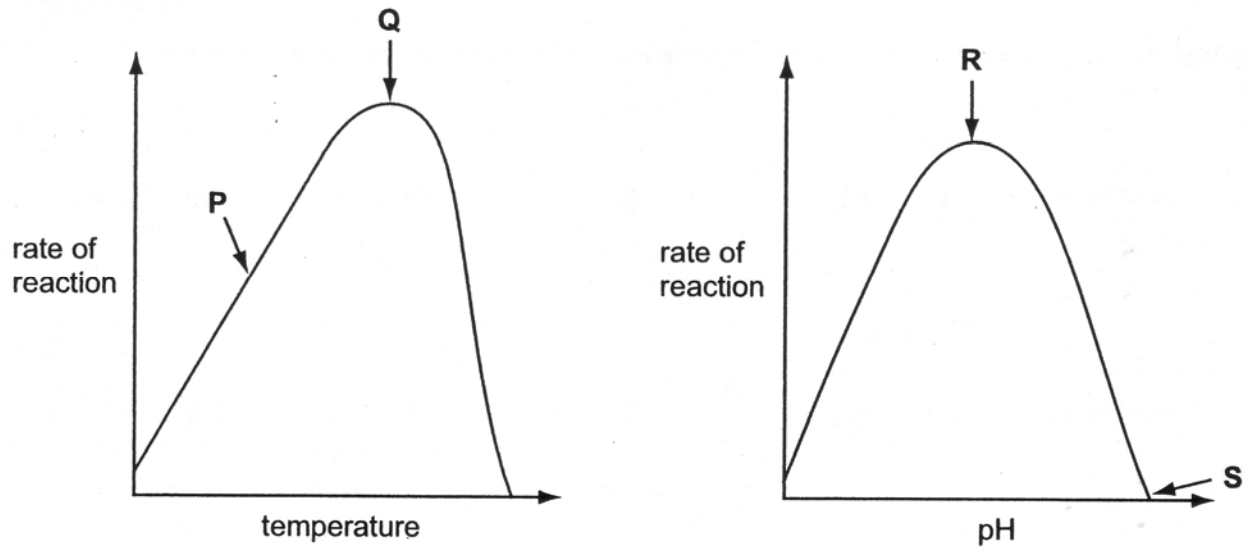
7. The graph shows the course of an enzyme-catalysed reaction at 30° C.



What is true at time X?

- A Most enzyme molecules will have free active sites.
- B The number of available substrate molecules is high.
- C The number of enzyme-substrate complexes is low.
- D The rate remains the same if more enzyme is added.

8. The graphs show the effects of temperature and pH on enzyme activity.



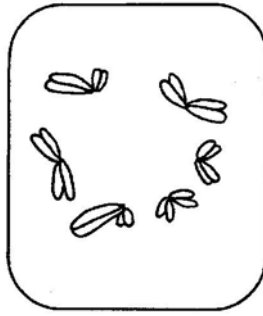
Which statement explains the enzyme activity at the point shown?

- A** At **P**, hydrogen bonds are formed between enzyme and substrate.
- B** At **Q**, the kinetic energy of enzyme and substrate is highest.
- C** At **R**, peptide bonds in the enzyme begin to break.
- D** At **S**, the substrate is completely denatured.

9. Which combination of chromosomes would result in a zygote showing polyploidy, if the parental chromosome number is 36?

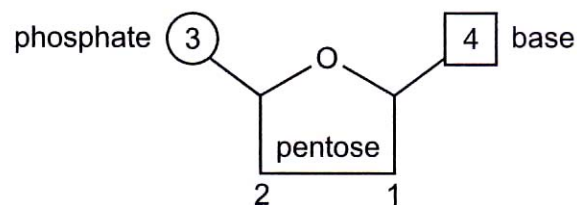
	Sperm	Ovum
A	9	9
B	18	18
C	18	19
D	18	36

10. What does this represent?



- A A cell during metaphase.
- B A stage of mitosis in a polyploidy cell.
- C Homologous chromosomes preparing to cross over.
- D Pairing of homologous chromosomes.

11. The diagram shows the structure of a nucleotide.



One type of polynucleotide consists of a single strand, twisted into a clover leaf shape. Which is the type of pentose of this molecule and the bonding between the nucleotides?

	Pentose	Bonding between nucleotides
A	deoxyribose	Between positions 1 and 3
B	deoxyribose	Between positions 2 and 3
C	ribose	Between positions 4 and 4
D	ribose	Between positions 2 and 3

12. One individual, organism **P**, in a sexually-reproducing population of mammals has two forms of a protein in its heart muscle. The normal form is identical to that in most other members of the population, while the variant form differs by one amino acid. The diagram shows the amino acid sequences in part of the normal protein and in the same part of the variant protein.

Normal protein	Variant protein
-gly-lys-lys-gly-glu-	-glu-lys-gly-gly-glu-

What could account for the presence of the normal protein and the variant protein in P?

- 1 The sequence of amino acids was altered in a cell in the heart muscle of P's mother.
- 2 The sequence of amino acids was altered in a cell in the ovaries of P's mother.
- 3 The sequence of bases was altered in the DNA in a cell in the heart muscle of P.
- 4 The sequence of bases was altered in the DNA in a cell in an ovary of P.
- 5 The sequence of bases was altered in the DNA in a cell in a testis of P's grandfather.

- A** 1, 2 and 4 only
- B** 1 and 3 only
- C** 2, 4 and 5 only
- D** 3 and 5 only

13. A peptide consists of ten amino acids of four different kinds. What is the theoretical minimum number of tRNA molecules required to translate the mRNA for this peptide?

- A** 4
- B** 10
- C** 12
- D** 30

14. In most organisms, all possible triplets of bases in DNA code for an amino acid, except for the coding strand 'stop' triples ATT, ATC and ACT. The coding strand is complementary to mRNA.

In the prokaryote, *Methanosarcina barkeri*, the coding strand triplet ATC codes for an unusual amino acid, pyrrolysine.

The **non-coding** strand of a section of DNA in *M. barkeri* has the following sequence of bases.

TTTTTATTGTATTACTAGTGTTAATGA

How many amino acids can be made into a peptide by *M. barkeri* from the **coding** strand of this region of DNA?

- A** 5 **B** 6 **C** 7 **D** 9

15. Which process in bacteria always allows chromosomal and non-chromosomal DNA to be transferred?

- A** Binary fission
- B** Conjugation
- C** Transduction
- D** Transformation

16. Drugs that are effective in treating viral infections

- A** induce the body to produce antibodies.
- B** inhibit the action of viral ribosomes.
- C** interfere with the synthesis of viral nucleic acid.
- D** change the cell-recognition sites on the host cell.

17. A virus has a base ratio of $(A + G) / (U + C) = 1$.
What type of virus is this?

- A** A single-stranded DNA virus.
- B** A single-stranded RNA virus.
- C** A double-stranded DNA virus.
- D** A double-stranded RNA virus.

18. A mutation that renders non-functional the product of a regulatory gene for an inducible operon would result in

- A** continuous transcription of the genes of the operon.
- B** complete blocking of the attachment of RNA polymerase to the promoter.
- C** irreversible binding of the repressor to the operator.
- D** negative control of transcription.

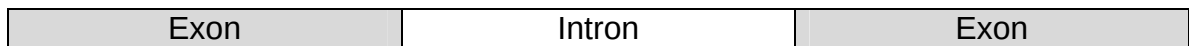
19. The human genome appears to have only two times the number of genes as the simple nematode, *C. elegans*. How can humans be so much more complex than nematodes?

- A** The unusually long introns in the human genes are involved in regulation of gene expression.
- B** More than one polypeptide can be produced from a gene by alternative splicing.
- C** Human genes code for many more types of domains.
- D** The human genome has a high proportion of noncoding DNA.

20. What describes the common features of the telomere and the centromere?

- A** Both areas have proteins associated with them.
- B** Both are made up of DNA with a high incidence of adenine and thymine.
- C** Both DNA sequences are conserved throughout the life of the organism.
- D** During interphase DNA replication takes place simultaneously in both regions.

21. The diagram shows a gene in a human cell.



In one form of the blood condition thalassaemia, there is a mutation in an intron of the haemoglobin gene.

Which stage of gene expression would this mutation directly affect?

- A** Activation of the gene by a steroid hormone
- B** Attachment of mRNA to a ribosome
- C** Binding of RNA polymerase to DNA
- D** Splicing of pre-mRNA

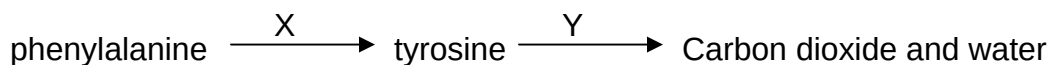
22. Which of the following is a difficulty in getting prokaryotic cells to express eukaryotic genes?

- A** The signals that control gene expression are different and prokaryotic promoter regions must be added to the vector.
- B** Prokaryotic cells cannot transcribe introns because their genes do not have them.
- C** The ribosomes of prokaryotes are not large enough to handle long eukaryotic genes.
- D** The RNA splicing enzymes of bacteria are different from those of eukaryotes.

23. Day-old chicks can be difficult to sex, but their down feathers reveal a sex-linked dominant character called 'barred'. Female birds are XY and males are XX. Which cross would ensure correct sexing of the resultant chicks?

	Female	Male
A	barred	homozygous barred
B	barred	homozygous unbarred
C	unbarred	heterozygous barred
D	unbarred	homozygous barred

24. The following reaction sequence occurs in humans.



Phenylketonuria (PKU) is a disease caused by an enzyme deficiency in step X, and alkaptonuria (AKU) is due to an enzyme deficiency in step Y. Both conditions are rare and caused by recessive alleles. A person with PKU marries a person with AKU. Which phenotypes would be expected for their children?

- A** All normal
- B** All AKU only
- C** All PKU only
- D** All both AKU and PKU

25. The table shows the blood group phenotypes resulting from crosses between different genotypes.

Genotype of 1 st parent	Genotype of 2 nd parent	
	$I^A I^A$	$I^A I^O$
$I^A I^O$	A	A and O
$I^B I^O$	A and AB	A, B, O and AB

Two parents have a son who has blood group A and phenylketonuria. One parent has blood group O and the other has blood group AB. Neither parent has phenylketonuria. What is the probability that the second child of these parents will be a girl with blood group B who does not have phenylketonuria?

- A 1 in 16
- B 1 in 8
- C 3 in 16
- D 3 in 8

26. The table summarises a breeding experiment with *Drosophila melanogaster*.

parents	BBNN	X	bbnn
F1	BbNn		
Test cross	BbNn	X	bbnn
Test cross offspring	BbNn	bbNn	Bbnn bbnn
No. of test cross offspring	586	111	106 465

Key

- B** wild type body colour
- b** black body colour
- N** normal wings
- n** vestigial wings

What do these results suggest?

- A Independent assortment has not occurred.
- B The alleles **N** and **n** are linked.
- C The alleles for normal and vestigial wings are codominant.
- D The gene loci for body colour and wing shape are not linked.

27. Which two products result from the photoactivation of chlorophyll within a chloroplast?

- A** ATP and reduced NADP
- B** ATP and H₂O
- C** CO₂ and ATP
- D** Reduced NADP and H₂O

28. Six tubes were set up as shown in the table.

tube	contents
1	Glucose + homogenized animal cells
2	Glucose + mitochondria
3	Glucose + cytoplasm lacking organelles
4	Pyruvate + homogenized animal cells
5	Pyruvate + mitochondria
6	Pyruvate + cytoplasm lacking organelles

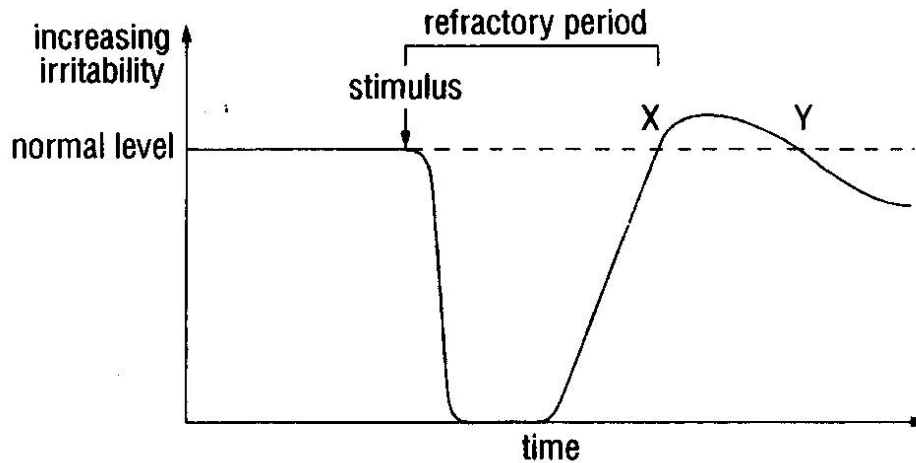
After incubation, each sample was analysed to determine the presence of carbon dioxide and lactate. In which tubes is lactate most likely to be present?

- A** 1 and 3 only
- B** 2, 3, 5 and 6 only
- C** 3 and 6 only
- D** 4, 5, and 6 only

29. Which substances enter and leave a mitochondrion during aerobic respiration?

	enters	leaves
A	acetyl Co-A	carbon dioxide
B	ADP	lactate
C	glucose	reduced NAD
D	pyruvate	ATP

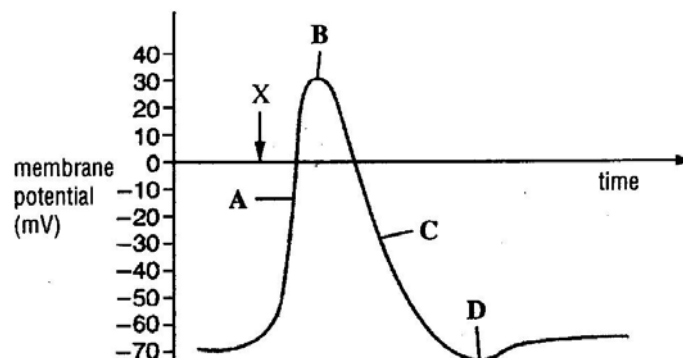
30. The figure shows the relative irritability of stimulated nerve fibre.



What would happen if a second stimulus is applied during the period XY?

- A A stimulus below the normal threshold will cause a response.
- B The fibre will respond repeatedly to a single stimulus.
- C The reaction to the stimulus will be greater than normal.
- D The reaction will occur more quickly than normal.

31. The graph shows the changes in membrane permeability when a stimulus X is applied to a neurone. Where is the point of maximum permeability to sodium ions?



32. Dinitrophenol is a metabolic poison that prevents the production of ATP. What will happen if a resting axon is treated with such a poison?

- A** The charge inside the axon membrane becomes more negative.
- B** The charge outside the axon membrane becomes less positive.
- C** The charge inside the axon membrane becomes less positive.
- D** The charge outside the axon membrane does not change.

33. Which of the following increases the number of different alleles in a population?

- A** Crossing over
- B** Gene mutation
- C** Random fusion of gametes
- D** Reassortment of chromosomes in meiosis

34. In a given phylum, which group contains the greatest number of species?

- A** class
- B** family
- C** genus
- D** order

35. Two populations of a given species could only evolve into two distinct species if they were subjected to

- A** disruptive selection.
- B** genetic isolation.
- C** geographical isolation.
- D** stabilising selection.

36. Which statements are true?

- 1 Continuous variation is always influenced by the environment.
- 2 Continuous variation is always caused by many genes.
- 3 Discontinuous variation is always caused by one gene.
- 4 Discontinuous variation is sometimes influenced by the environment.

A 1 and 4

B 2 and 3

C 2 and 4

D 3 and 4

37. Some statements concerning evolution are listed:

1. Characteristics acquired during an organism's lifetime are passed to its offspring.
2. Individuals in a sexually reproducing population are different.
3. More offspring are produced than can possibly survive to sexual maturity.
4. Offspring tend to resemble their parents.
5. The fossil record shows that many species have become extinct.

Which of these statements form the basis of Darwin's theory of evolution by natural selection?

A 1 3 4

B 1 4 5

C 2 3 4

D 2 4 5

38. If the first three nucleotides in a six-nucleotide restriction site are CTG, what would the next three nucleotides most likely be?

A AGG

B GTC

C CTG

D CAG

39. At the start of the polymerase chain reaction (PCR), single stranded primers are added to the denatured DNA and the mixture cooled to 60°C.

What explains why the denatured DNA strands anneal with primers and **not** each other?

- A** The primers are shorter and anneal more easily.
- B** The primers anneal only to the 3' end of denatured DNA.
- C** The primer concentration exceeds the denatured DNA concentration.
- D** The temperature prevents the denatured DNA from annealing together.

40. Which of the following technique does not involve any nucleic acid hybridization?

- A** Gel electrophoresis
- B** Southern blotting
- C** Polymerase Chain Reaction
- D** DNA fingerprinting

Answers for H2 P1 Anderson Junior College Biology Prelims

1	C	9	D	17	D	25	C	33	B
2	B	10	A	18	A	26	A	34	A
3	B	11	D	19	B	27	A	35	B
4	A	12	D	20	A	28	C	36	C
5	D	13	A	21	D	29	D	37	C
6	B	14	C	22	A	30	A	38	D
7	B	15	A	23	B	31	A	39	C
8	A	16	C	24	A	32	B	40	A